

Sequences and series



- 1 Describe the difference between a sequence and a series.
- 2 Find the third term in the sequence $4, -5, 6, -7, 8$.
- 3 Find the next number in the sequence $16, 36, 64, 100, 144, \dots$
- 4 Consider the sequence a_n . If the first term is a_1 , write the notation used for the fifth term of the sequence.
- 5 Determine whether each sequence is finite or infinite:
 - a $1, 2, 3, 4, 5, 6$
 - b $-1, -2, -3, -4, -5, -6, \dots$
 - c $-1, -2, -3, -4, -5$
 - d $1, 2, 3, 4, 5, 6, \dots$
- 6 If all the dates in July were used to form a sequence, would the sequence be finite or infinite?
- 7 Identify the following sequences as finite or infinite:
 - a $a_1 = 2; a_n = 2a_{n-1}$ for $n \geq 2$
 - b $a_2 = 3; a_n = a_{n-1} + a_{n-2}$ for $3 \leq n \leq 9$
- 8 State the first five terms of the following sequences. Round terms to four decimal places where necessary.
 - a $a_n = 3n - 3$
 - b $a_n = -2^n$
 - c $a_n = 2^{n+1}$
 - d $a_n = (-2)^n (3n)$
 - e $a_n = \frac{1}{\sqrt{n}}$
 - f $a_n = \left(1 + \frac{1}{n}\right)^n$
- 9 State the first four terms, in exact form, of the following sequences:
 - a $a_n = \frac{n+4}{n+3}$
 - b $a_n = \frac{5n-1}{n^2+5}$
 - c $a_n = \left(\frac{1}{2}\right)^n (n-3)$
 - d $a_n = (-1)^{n-3} (n-1)$

10 For each of the following sequences, state the first four terms:



- a $a_1 = 4, a_n = na_{n-1} - 5$ for $n > 1$
- b $a_1 = 3, a_n = a_{n-1} + n$ for $n > 1$
- c $a_1 = -2, a_n = 2n + 3a_{n-1}$ for $n > 1$
- d $a_1 = 3, a_2 = 5, a_n = a_{n-1} \times a_{n-2}$ for $n > 2$
- e $a_1 = 3, a_2 = 2, a_n = 3(a_{n-1})^2 - 4a_{n-2}$ for $n > 2$

Sigma notation

11 Write the following series using sigma notation:

- a $4 + 8 + 12 + 16 + 20 + \dots$
- b $-3 + 9 - 27 + 81 - 243 + 729$
- c $\frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3}$
- d $\frac{1}{3 \times 1} + \frac{1}{3 \times 2} + \frac{1}{3 \times 3} + \dots + \frac{1}{3 \times 7}$
- e $\frac{2}{5+1} + \frac{2}{5+2} + \frac{2}{5+3} + \dots + \frac{2}{5+11}$
- f $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots + \frac{1}{1024}$

12 Find the value of the following:

- a $\sum_{r=1}^5 (4r - 4)$
- b $\sum_{k=3}^5 \frac{1}{k}$
- c $\sum_{r=1}^4 \frac{1}{r+2}$
- d $\sum_{k=1}^5 (-1)^{k-1} k^2$
- e $\sum_{r=-2}^1 (3r^2 - 1)$
- f $\sum_{n=1}^5 5(3)^n$
- g $\sum_{r=1}^5 (3^r + 1)$
- h $\sum_{k=0}^3 ((-3)^k + 1)$
- i $\sum_{n=1}^5 n^{n-3}$
- j $\sum_{n=-3}^0 (n^2 - 4n)$

13 Given that $x_1 = 2, x_2 = 3, x_3 = 4, x_4 = 5$ and $x_5 = 6$, find the value of $\sum_{n=1}^5 x_n$.

14 Given that $x_1 = 1, x_2 = 0, x_3 = -1, x_4 = -2$ and $x_5 = -3$, find the value of $\sum_{j=1}^5 2x_j$.

15 Given that $x_1 = -1, x_2 = 0, x_3 = 1, x_4 = 2$ and $x_5 = 3$, find the value of $\sum_{r=1}^5 (4x_r + 2)$.



17 Given that $x_1 = 2$, $x_2 = 1$, $x_3 = 0$, $x_4 = -1$ and $x_5 = -2$, find the value of the following:

a $\sum_{i=1}^4 (x_i^2 - 3)$

b $\sum_{n=2}^4 \left(\frac{x_n - 4}{x_n + 2} \right)$

18 Consider $f(x) = 4x + 1$, $x_1 = -1$, $x_2 = -2$, $x_3 = -3$, $x_4 = -4$ and $\delta x = \frac{3}{2}$.

a $\sum_{k=1}^4 f(x_k) \delta x$

b $\sum_{x=1}^5 f(x)$

19 Consider the series $\sum_{i=1}^7 (i + i^3)$ and $\sum_{i=1}^7 i + \sum_{i=1}^7 i^3$.

a Write each series in expanded form.

b Do the two sums have the same value?

c Is $\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$?

Applications

20 A doctor administered pain medication in the following amounts in milligrams over four days after their patient underwent surgery:

Year (n)	1	2	3	4
Dosage (m)	2000	1520	1155.2	878.0

Find the recursive formula which models this sequence. Round the values to two decimal places if necessary.

21 Christa deposits \$1 into a new bank account while Jenny deposits \$38 into a new bank account. The next day, Christa adds \$2 to the account, and doubles how much she adds each day thereafter. Jenny adds \$5 each day.

a List the amount Christa had in her account each day for the first four days.

b List the amount Jenny had in her account each day for the first four days.

c On which day will they have the same amount of money in their bank accounts?